CAREBRIDGE AI POST-DISCHARGE CARE COORDINATION

PROJECT PROFILE / 2026

Care, connected. At every step.



A patient and caregiver workspace for organizing post-discharge instructions, tasks, reports, and access barriers.

One shared journey
Tasks, appointments, and follow-up

Patient-reviewed AI
Drafts stay under patient control

Support with boundaries
Caregiver access follows assignment

Project overview Product, architecture, workflows, safeguards, and delivery notes. Working MVP prepared October 30, 2026.

01 / EXECUTIVE SUMMARY

What the project does

CareBridge AI | Professional preview | Project demo information only

CareBridge AI is a web application for patients and caregivers coordinating the practical work that follows a hospital visit. It brings healthcare journey tasks, appointments, reports, documents, caregiver assignments, notifications, and access barriers into one workspace.

> **Clinical boundary:** CareBridge AI is an organization and coordination tool. It does not diagnose, prescribe, or change a clinician's instructions. AI-generated extraction is a draft for patient review, and clinical wording is not editable by caregivers.

For patients

See the journey at a glance, track due and completed tasks, upload and review documents, manage reports, invite caregivers, assign selected tasks, and record barriers.

For caregivers

Accept an invitation, view tasks explicitly assigned to them, update task status, and see assignment history. Patient-only records and unassigned work remain outside their view.

For the care journey

Keep instructions connected to their source document, distinguish administrative progress from medical readiness, and make follow-up work easier to coordinate.

For deployment

Run locally with SQLite, or configure PostgreSQL for a hosted service. The React client and FastAPI API can be deployed separately.

Product principles

- **Patient review comes first:** uploaded instructions do not silently become confirmed tasks.
- **Access follows relationships:** caregivers see work only after invitation and assignment.
- **Administrative readiness stays administrative:** a report checklist is not a clinical clearance indicator.
- **Fallbacks are explicit:** without an AI provider key, analysis is labeled simulated and does not invent instructions.

02 / USERS AND EXPERIENCE

Roles and main screens

Role	Capabilities	Access boundary
Patient	Own profile, dashboard, journey, tasks, document uploads and review, report checklist, caregiver links, barriers, and notifications.	Can manage their own records and choose which journey tasks to share.
Caregiver	Assigned tasks, status updates, and task assignment history in a shared journey.	Must be linked through an invitation; cannot edit clinical task wording or access another patient's records.
Demo user	Fictional patient and caregiver accounts with sample journey content.	Sample data only; do not enter real patient information into the demo workspace.

Dashboard

Journey progress, task status, overdue administrative work, upcoming appointments, report availability, recent notifications, and caregiver activity.

Healthcare journey

A timeline of tasks with dates and statuses, supporting a simple view of the next steps after discharge.

Documents

Upload supported files, inspect extracted details, provide consent for AI analysis, review the draft, and explicitly confirm or reject suggested actions.

Tasks and reports

Create and update tasks, view task history, track required reports, and associate verified reports with supporting documents.

Caregivers

Invite caregivers, assign or remove work, and review who has responsibility for a selected task.

Barriers and notifications

Record an access barrier and follow-up notes; review reminders and mark notifications as read.

Typical patient flow

1. Sign in Patient opens their workspace. > **2. Add source** Upload a discharge or care document. > **3. Review draft** Inspect extracted details and task suggestions. > **4. Confirm** Patient approves supported items. > **5. Coordinate** Track work and share selected tasks.

The interface has routes for Dashboard, Journey, Documents, Tasks, Reports, Caregivers, Barriers, Notifications, and Settings.

03 / CAPABILITIES

Functional scope

CareBridge AI Project Overview — Functional scope information only

Area	What is implemented
Identity and profiles	Patient and caregiver registration, login, authenticated profile retrieval, JWT access tokens, and patient profile management.
Journey and tasks	Create, view, edit, complete, delete, assign, and track tasks. Task history records caregiver assignment changes. Overdue work is surfaced in the dashboard.
Documents	PDF, JPEG, and PNG upload with content type, file signature, ownership, and size validation. Default maximum size is 10 MB. Preview requires an authenticated owner.
Extraction and confirmation	PDF text extraction and image OCR adapter; structured AI extraction can run through an OpenAI-compatible provider. Analysis requires consent. Suggestions remain unverified until patient confirmation.
Reports	Track required reports and their availability. Verification requires a supporting document. Readiness reflects document availability only.
Caregiver sharing	Invitation-based relationships and task-level assignments. Caregivers can update task status but cannot edit clinical task instructions.
Notifications	In-app reminders with deduplication and per-user visibility; external email, SMS, and WhatsApp delivery are not implemented.
Access barriers	Create, view, and update operational barriers that affect access to care, including status and follow-up notes.

Document review safeguards

1. The user uploads a supported file. The API checks extension, MIME type, file signature, size, and ownership.
2. Text extraction and analysis produce a reviewable record. AI analysis is not called until the patient has provided explicit consent.
3. Proposed instructions are shown with their source context. Confirmation checks that task wording is supported by the source content.
4. Confirmed suggestions are still labeled as needing verification where a clinician should review them.

> **OCR boundary:** text-based PDFs are supported through PyMuPDF. Image OCR requires the Python Tesseract adapter and a system Tesseract executable. Scanned PDFs currently return an actionable unsupported-OCR response.

04 / TECHNICAL ARCHITECTURE

How the application is built

CareBridge AI | Project overview | Fictional demo information only

Frontend

React 18 with Vite, React Router, Axios, Tailwind CSS, Recharts, and Lucide icons. The frontend is a single-page application with responsive views and protected patient/caregiver routes.

Backend

FastAPI exposes JSON REST endpoints. SQLAlchemy handles persistence; Pydantic schemas validate API data. Alembic migrations support controlled schema updates.

Database

SQLite is the local default. PostgreSQL is configured through `DATABASE_URL`; hosted deployment is described around Render and a managed PostgreSQL database.

AI and OCR

AI provider settings remain server-side. Structured extraction uses an OpenAI-compatible endpoint. PDF text extraction and optional image OCR are separate adapters.

Request flow

React client Page action or data request. > **Axios API client** Base URL and bearer token. > **FastAPI route** Authentication and validation. > **Service and ORM** Rules, audit, and database work. > **JSON response** Client updates the page.

Core data records

Record family	Purpose
Users and profiles	Identity, role, demo flag, and patient-specific contact/profile fields.
Caregiver relationships	Invitation, linked caregiver, and journey-sharing relationship.
Documents and extracted data	Uploaded file metadata, extracted text, analysis result, consent, and confirmation state.
Healthcare journeys and tasks	Patient journey, due dates, statuses, source document references, assignments, and history.
Reports and notifications	Required report checklist, document association, notification ownership, and reminder dedupe key.
Barriers and audit logs	Access-related follow-up records and a record of sensitive application actions.

API surface

Main route groups include `/api/auth`, `/api/patients`, `/api/dashboard`, `/api/journey`, `/api/tasks`, `/api/documents`, `/api/reports`, `/api/caregivers`, `/api/notifications`, and `/api/barriers`. The service root and `/api/health` provide API

information and health status.

05 / SECURITY AND RESPONSIBLE USE

Access, privacy, and clinical boundaries

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Authentication

Passwords are stored as scrypt hashes. API sessions use signed, expiring JWTs. The secret key must be replaced with a strong value outside demo mode.

Authorization

Patient routes are scoped to the signed-in patient's records. Caregiver access requires an invitation and task assignment. Notification and document access are scoped to the owner.

File access

Uploads have type, signature, and size checks. Stored files are not exposed through a public URL; previews require authenticated ownership checks.

AI handling

Provider credentials stay on the backend. Analysis requires consent and remains a draft. When no key is set, simulated mode is clearly labeled and does not generate medical tasks.

Information that should not be inferred

- A completed report checklist does not mean a patient is medically ready for treatment or discharge.
- A model-generated explanation is not a diagnosis, recommendation, prescription, or substitute for a clinician.
- In-app notifications do not imply an email, SMS, or messaging service sent an alert.

Production readiness: the included workspace is an MVP and uses fictional sample data. Before real patient use, configure private durable file storage, encryption, access controls, retention and deletion rules, monitoring, backups, provider terms, and applicable privacy/legal review. Do not use the demo workspace for real patient information.

06 / RUNNING AND DEPLOYING

Development and delivery

Local setup

1. Use Python 3.11 or newer and Node.js 18 or newer.
2. In `backend/`, create and activate a virtual environment, install `requirements.txt`, and copy `.env.example` to `.env`.
3. Start the API with `uvicorn app.main:app --reload --port 8000`. It serves API information at `http://localhost:8000/`, health at `/api/health`, and API docs at `/docs`.
4. In `frontend/`, install npm dependencies and run `npm run dev`. The client is served at `http://localhost:5173` and calls the API at `http://localhost:8000/api`.

Configuration

Setting	Role
`DATABASE_URL`	SQLite by default; PostgreSQL URL for hosted use.
`SECRET_KEY`	JWT signing secret; use a strong generated secret outside demo mode.
`DEMO_MODE`	Seeds fictional accounts and sample journey content when enabled.
`CORS_ORIGINS`	Browser origins permitted to call the API. Local demo mode supports localhost and 127.0.0.1 on port 5173.
`AI_API_KEY / AI_MODEL / AI_BASE_URL`	Optional server-side structured extraction provider configuration.
`UPLOAD_DIR / MAX_UPLOAD_MB`	Local upload storage path and size ceiling.
`VITE_API_URL`	Frontend API base URL, including the `/api` suffix.

Deployment model

The root Render blueprint defines the Python API service, a PostgreSQL database, and a pre-deploy Alembic migration. The React client can be deployed to Vercel with its SPA rewrite and a `VITE_API_URL` pointing at the deployed API. The backend CORS list must include the deployed frontend origin.

Uploaded documents use local storage by default. A production deployment should use private durable object storage and verify backup, retention, restore, and access-control behavior before enabling real document uploads.

07 / VERIFICATION AND ROADMAP

Current status

CareBridge AI project overview. Fictional demo information only

Verification	Result
Backend API suite	14 tests passed, including authentication, patient scoping, caregiver assignments, upload validation, report verification, simulated AI, CORS configuration, and legacy SQLite schema migration.
Frontend unit suite	2 tests passed for dashboard calculations.
Production frontend build	Vite build completed successfully.
Local HTTP smoke check	Frontend and its modules returned successfully. Login, dashboard, journey, tasks, report readiness, notifications, caregivers, and barriers returned HTTP 200. CORS preflight passed for both local hostnames.
Not exercised in this environment	Hosted PostgreSQL/Alembic deployment, live AI provider response, and actual Tesseract OCR processing.

Practical next steps

1. Choose and document a production privacy and data-retention model before onboarding real users.
2. Replace local file storage with private durable object storage and verify authorization for every file operation.
3. Exercise Alembic migrations against a managed PostgreSQL staging database and test backup restoration.
4. Run provider-specific AI evaluation using representative, de-identified documents and review false-positive and missed-instruction behavior.
5. Install and validate Tesseract for supported image workflows, or define an explicit scanned-PDF OCR path.
6. Add operational monitoring, rate limiting, account recovery, and production deployment checks as appropriate.

> **Project summary:** CareBridge AI provides a working foundation for organizing post-discharge coordination. Its strongest product boundary is the separation between clinician source material, AI-assisted drafts, and patient-confirmed tasks. Production use depends on completing privacy, storage, operations, and clinical workflow validation.

Repository structure: `backend/` contains the FastAPI API, database models, services, migrations, and tests; `frontend/` contains the React dashboard and pages. See `README.md` for the current setup instructions.